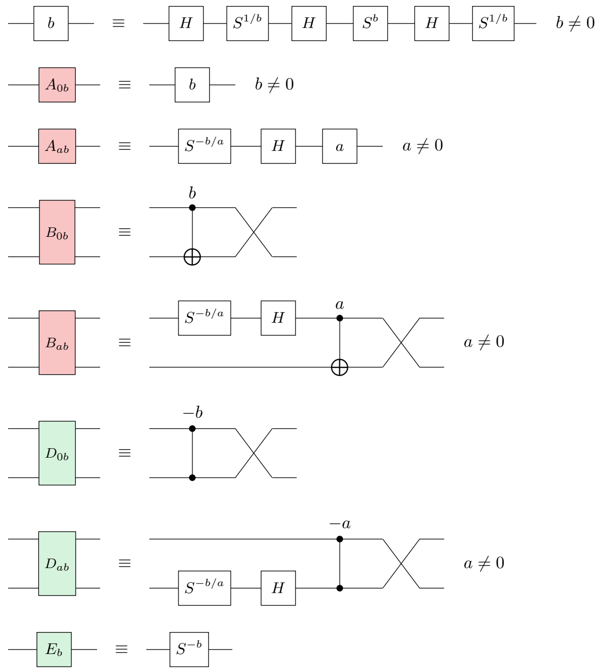
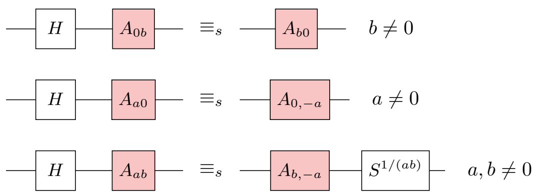
Box Relations

Box definitions



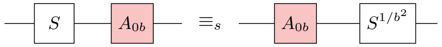
$A \leftarrow H$: pushing H through an A box

$$[x]^a \cdot H \;\equiv_s\; \mathrm{dir} \cdot [x']^a$$

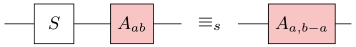


$A \leftarrow S$: pushing S through an A box

$$[x]^a \cdot S \;\equiv_s\; \mathrm{dir} \cdot [x']^a$$



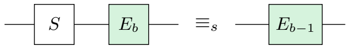
$b \neq 0$



$a \neq 0$

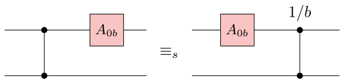
$E \leftarrow S$: pushing S through an E box

$$[b]^e \cdot S \;\equiv_s\; [b-1]^e$$

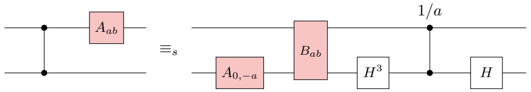


$L \leftarrow CZ$: pushing CZ through an L box

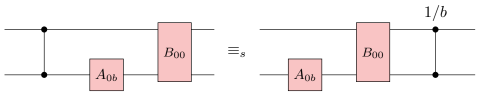
$$[l]^\ell \cdot CZ \;\equiv_s\; \mathrm{dir} \cdot [l']^\ell$$



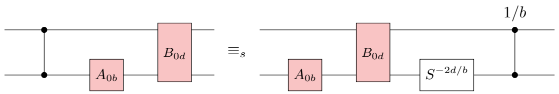
$b \neq 0$



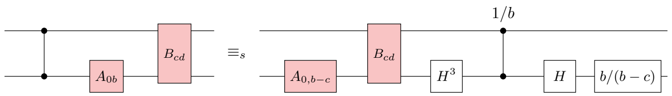
$a \neq 0$



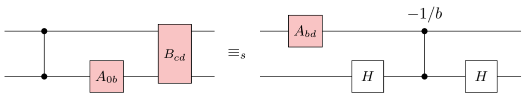
$b \neq 0$



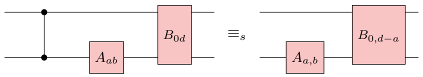
$b, d \neq 0$



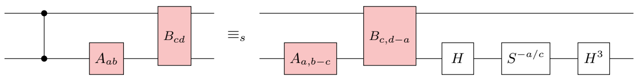
$b, c, b-c \neq 0$



$b = c \neq 0$



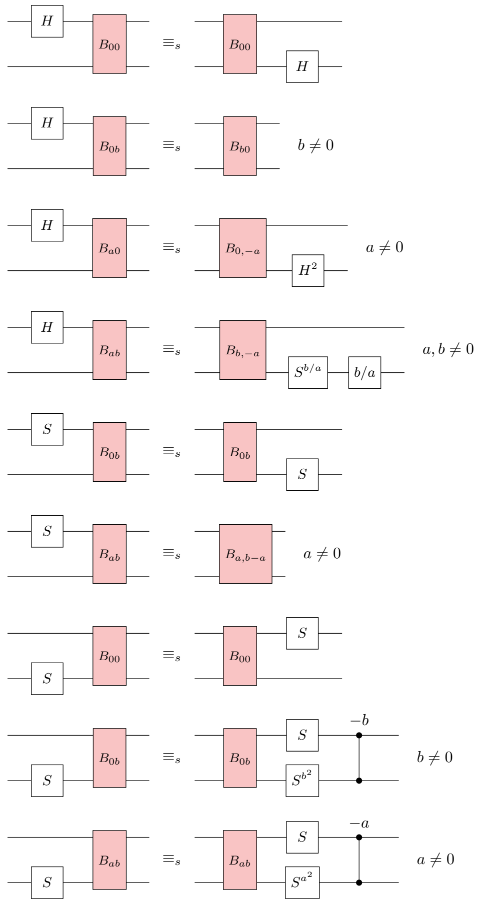
$a \neq 0$



$a, c \neq 0$

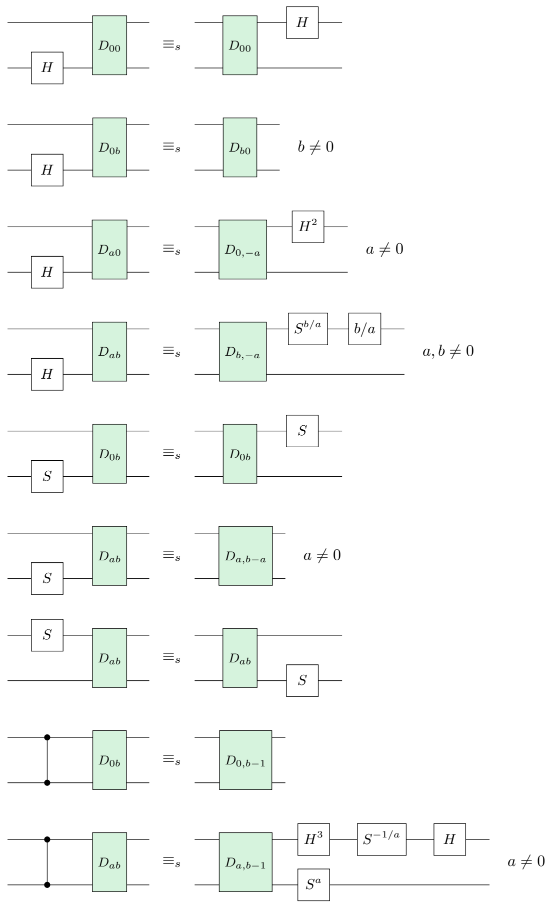
$B \leftarrow H^{\uparrow}, S^{\uparrow}, S$: pushing $H^{\uparrow}$, $S^{\uparrow}$, S through a B box

$$[b]^b \cdot [g]^w \;\equiv_s\; \mathrm{dir} \cdot [b']^b$$



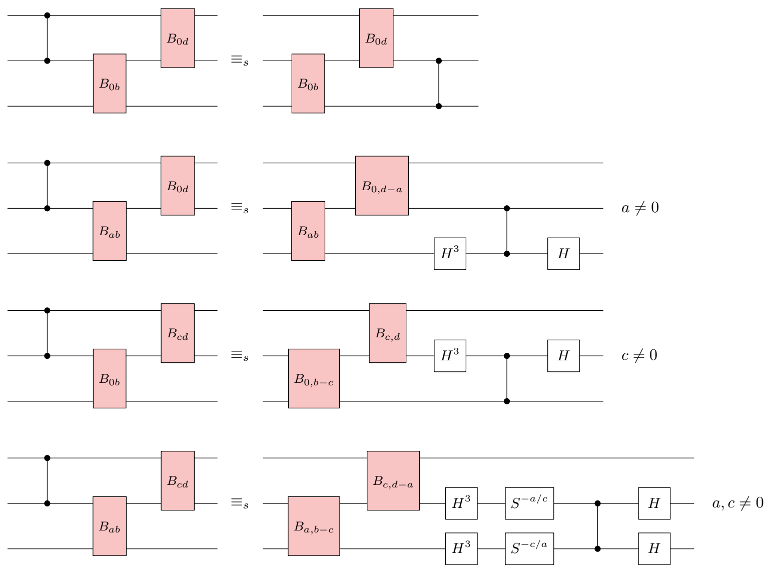
$D \leftarrow H, S^{\uparrow}, S, CZ$: pushing H, $S^{\uparrow}$, S, CZ through a D box

$$[d]^d \cdot [g]^w \;\equiv_s\; S^e \cdot \mathrm{dir}^{\uparrow} \cdot [d']^d$$



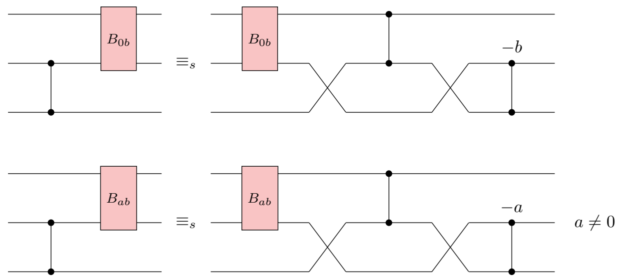
$BB \leftarrow CZ^{\uparrow}$: pushing $CZ^{\uparrow}$ through a BB box

$$[vb]^{vb} \cdot CZ^{\uparrow} \;\equiv_s\; \mathrm{dir}\Downarrow \cdot [vb']^{vb}$$



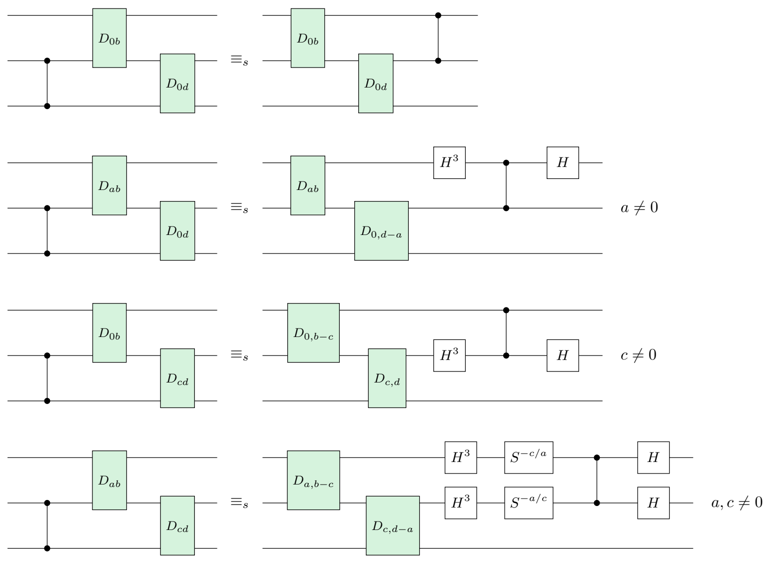
$B^{\uparrow} \leftarrow CZ$: pushing CZ through a $B^{\uparrow}$ box

$$[b]^b{\uparrow} \cdot CZ \;\equiv_s\; \mathrm{dir} \cdot [b]^b{\uparrow}$$



$DD \leftarrow CZ$: pushing CZ through a DD box

$$[vd]^{vd} \cdot CZ \;\equiv_s\; \mathrm{dir}^{\uparrow} \cdot [vd']^{vd}$$


D_{0b} D_{0d} $\equiv_s$ D_{0b} D_{0d}
D_{ab} D_{0d} $\equiv_s$ D_{ab} $D_{0,d-a}$ H^3 H $a \neq 0$
D_{0b} D_{cd} $\equiv_s$ $D_{0,b-c}$ $D_{c,d}$ H^3 H $c \neq 0$
D_{ab} D_{cd} $\equiv_s$ $D_{a,b-c}$ $D_{c,d-a}$ H^3 H^3 $S^{-c/a}$ $S^{-a/c}$ H H $a, c \neq 0$